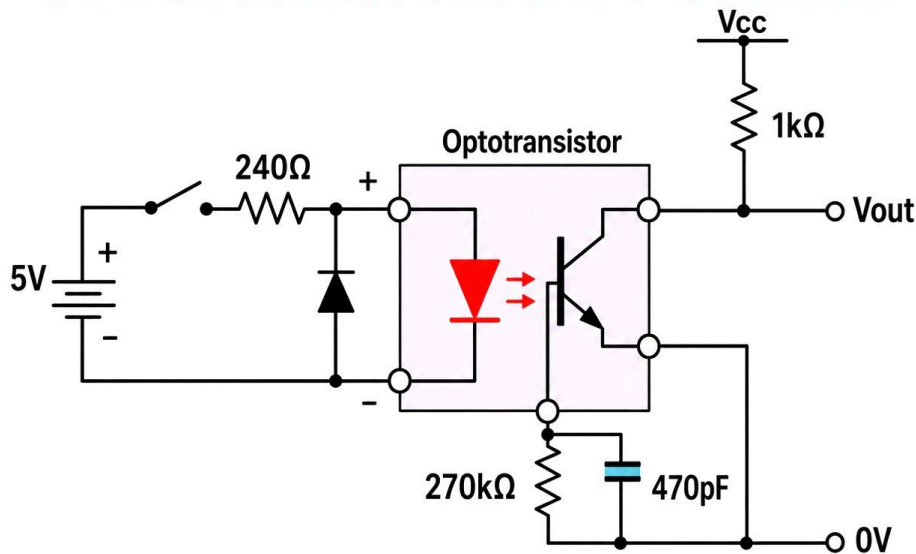




OPTOTRANSISTOR DC SWITCH



TYPICAL APPLICATIONS:

- 1. Isolated Relay Driving:** Controlling a high-current relay with complete input protection.
- 2. Signal Isolation:** Passing digital signals between circuits with different ground potentials.



Electronics Education

8 maj kl. 16:46 · 🌐

An optotransistor DC switch is used to transfer a signal between two circuits while keeping them electrically isolated. Inside the optocoupler, an LED and a phototransistor work together using light instead of a direct electrical connection.

When the input switch is turned ON, current flows through the 240Ω resistor and lights the internal LED. The light activates the phototransistor on the output side. This allows current to flow through the transistor, pulling the output voltage toward 0V.

When the input switch is OFF, the LED turns OFF, the phototransistor stops conducting, and the 1kΩ pull-up resistor pulls the output back to Vcc. This creates a clean digital switching action.

The isolation between input and output helps protect sensitive circuits from voltage spikes, electrical noise, and ground loop problems. Even though the two sides exchange signals, they do not share a direct electrical path.

The 270kΩ resistor improves transistor stability, while the 470pF capacitor helps reduce noise and switching glitches in fast circuits.

Common applications include relay control, microcontroller interfacing, industrial automation, power supply feedback circuits, and safe signal transfer between different voltage systems.

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